Finite Groups and Their Representations — Exercise Sheet 9

Character arithmetic and invariant forms

Variant: Questions only

Exercise 1 (An average of algebraic integers). Let $\zeta = \exp(2\pi i/3)$ and $\alpha = (1+\zeta)/2$. Find the monic minimal polynomial of α over $\mathbb{Q}$ and prove that α is not an algebraic integer. Use the companion-matrix construction of Lecture 9 to realise α as an eigenvalue of a rational matrix. Explain why this does not contradict closure of algebraic integers under addition.

Exercise 2 (The coprime-degree step in S_4). Let $g = (1\,2)(3\,4)$ in S_4, and let $\mathcal{K} = [g]$. Recall from Lecture 6 that the irreducibles are $\mathbf{1}$, sgn, W, sgn $\otimes\, W$ and the lift of E through $S_4 \to S_3$, written F on this sheet, of dimensions $1, 1, 3, 3, 2$. Here W is the zero-sum subspace of $\mathbb{C}^4$, and F is the zero-sum plane in the permutation action on the three pairings of four points. Find every non-trivial irreducible U with degree coprime to $|\mathcal{K}|$ and $\chi_U(g) \neq 0$. Compute $\omega_U(\widehat{\mathcal{K}})$ and the scalar $\rho_U(g)$ for these U. Find their kernels. Verify column orthogonality between e and g. For F, exhibit the Bézout step proving that $\chi_F(g)/\dim F$ is an algebraic integer.

Exercise 3 (The character table of A_5). Use the classes $e, 2A, 3A, 5A, 5B$, represented by $e, (1\,2)(3\,4), (1\,2\,3), c, c^2$, where $c = (1\,2\,3\,4\,5)$. Their sizes are $1, 15, 20, 12, 12$ by Lectures 3–4. Let U_4 be the zero-sum subspace of the natural permutation module $\mathbb{C}^5$.

(i) Show that every class is inverse-stable and that squaring exchanges $5A$ and $5B$.

(ii) Compute χ_{U_4} and the characters of $\mathrm{Sym}^2 U_4$ and $\Lambda^2 U_4$. Remove the trivial module and U_4 from $\mathrm{Sym}^2 U_4$. Prove that the resulting module U_5 is irreducible.

(iii) Use $\Lambda^2 U_4$, row orthogonality, column orthogonality and the degree-square identity to complete the character table. Denote the remaining two modules by U_3^+ and U_3^-, choosing the larger value at c for U_3^+.

Exercise 4 (Galois conjugates of characters). Let G be finite, let $m = \mathrm{lcm}\{o(g) : g \in G\}$ be its exponent, so $g^m = e$ for every $g \in G$, and let γ be an automorphism of $\mathbb{Q}(\exp(2\pi i/m))$. You may use the field-theory fact that γ extends to an automorphism $\widetilde{\gamma}$ of $\mathbb{C}$. For an irreducible representation $\rho\colon G \to \mathrm{GL}(U)$, apply $\widetilde{\gamma}$ to every matrix entry in a fixed basis to define $\rho^{\widetilde{\gamma}}(g)$. Write U^γ for the resulting representation. Prove that the resulting representation is irreducible, has character $\gamma \circ \chi_U$, and has the same Frobenius–Schur indicator as U. For A_5, whose exponent is 30, you may also use that, with $\zeta_5 = \exp(2\pi i/5)$, $\sqrt{5} = 1 + 2(\zeta_5 + \zeta_5^{-1})$ lies in $\mathbb{Q}(\exp(2\pi i/30))$ and that some automorphism of this field sends $\sqrt{5}$ to $-\sqrt{5}$. Apply the result to any such automorphism and the table of Exercise 3.

Exercise 5 (Invariant forms for A_5). Using the table and module names in Exercise 3, compute the Frobenius–Schur indicator of every irreducible A_5-module. State what the theorem of Lecture 9 proves about their invariant complex bilinear forms. Include the square map on the five conjugacy classes.

Exercise 6 (Real forms of the dihedral planes). For $n \geq 3$, take $D_n = \langle r, s \rangle$ and the irreducible

planes of Sheet 5. Write U_k for the plane with generator matrices in an eigenbasis

$$R_k = \begin{pmatrix} \zeta^k & 0 \\ 0 & \zeta^{-k} \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \zeta = \exp(2\pi \mathrm{i}/n), \quad 2k \not\equiv 0 \pmod{n}.$$

Compute their indicators for all these parameters. Exhibit a basis in which both matrices are real, and a non-degenerate invariant symmetric bilinear form in the eigenbasis. Include the linear irreducibles when listing indicators.

Exercise 7 (⋆ A stronger degree divisibility). Let U be an irreducible representation of a finite group G, of degree d. Prove that d divides $[G : Z(G)]$, using the degree-divides-order theorem of Lecture 9 and the following construction. For every integer $q \geq 1$, form the external product $U^{\boxtimes q}$ for G^q. Its underlying tensor space has the action

$$(g_1, \ldots, g_q)(u_1 \otimes \cdots \otimes u_q) = g_1 u_1 \otimes \cdots \otimes g_q u_q.$$

Prove this representation is irreducible. Factor it through the normal subgroup

$$N_q = \{(z_1, \ldots, z_q) \in Z(G)^q : z_1 \cdots z_q = e\}.$$

Apply degree divisibility in the quotient for arbitrarily large q, and compare prime exponents. This optional argument is independent of the core exercises.